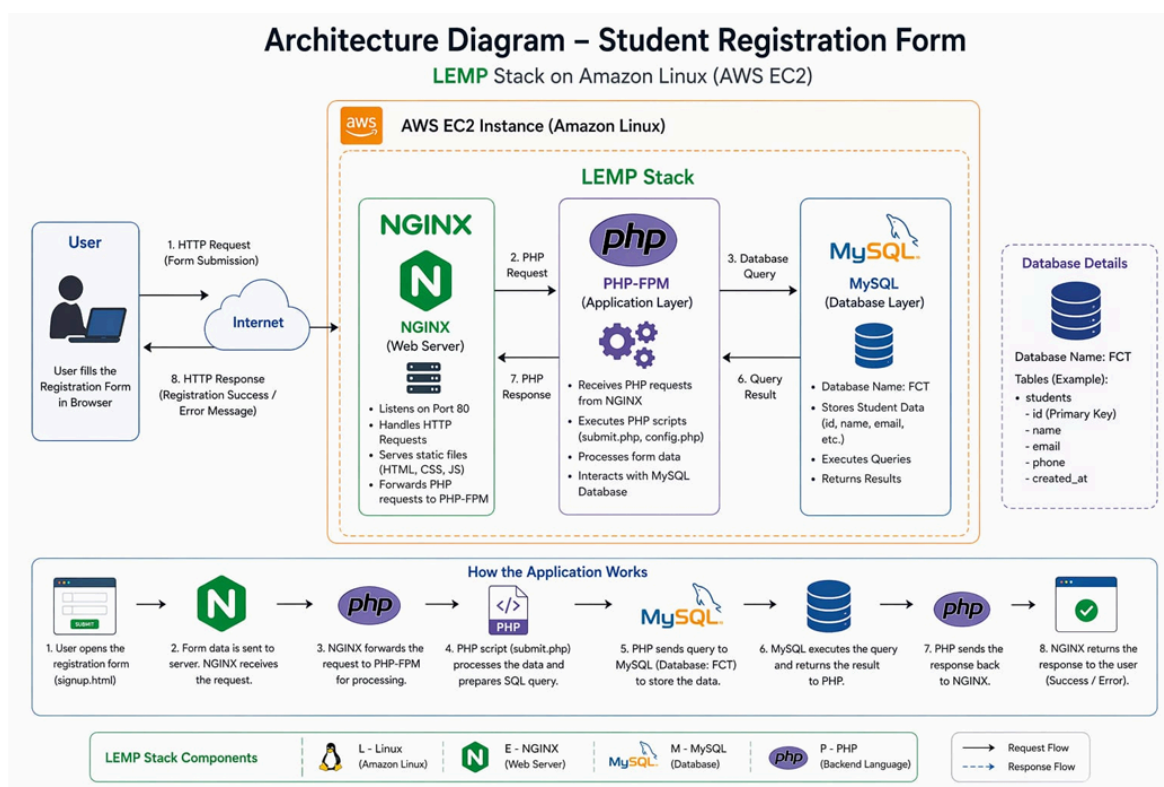


Registration Form Deployment (Dynamic Website LEMP Stack)

1)Introduction:

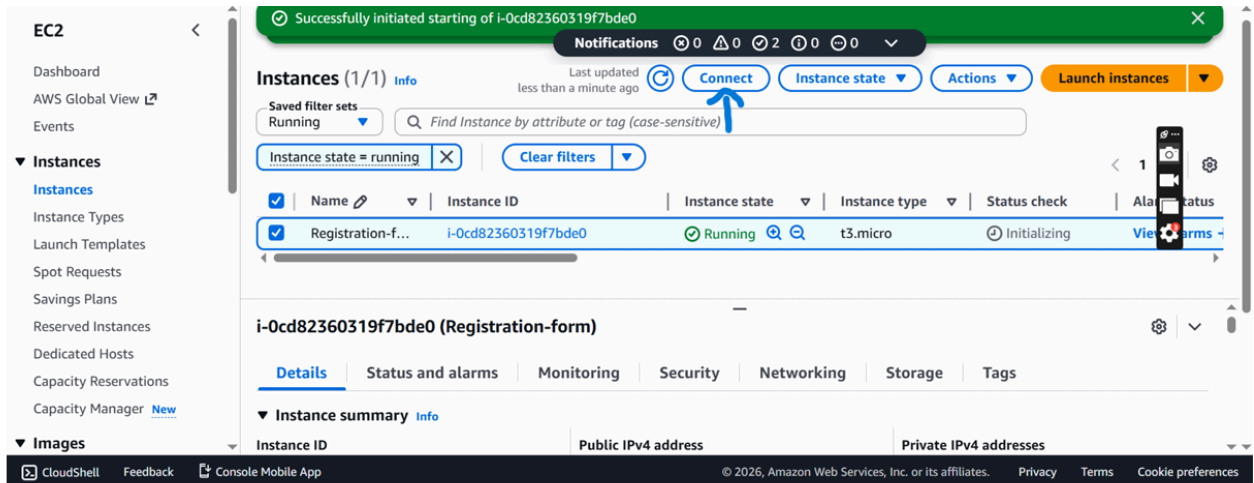
This is my first dynamic website called Student Registration Form. In this Website where users can enter their details like name, email, and other information. The website is built using: Frontend → HTML , CSS Backend → PHP Database → MySQL It is deployed on an Amazon Linux EC2 server using the LEMP stack (Linux, NGINX, MySQL, PHP). This setup helps the website to run online and store student data securely.

2) Architecture Diagram:



3)Deployment :

1:create EC2 instance



2:Install LEMP : sudo yum update

```
sudo yum install nginx php php-fpm
```

```
php-mysqlnd mariadb-server -y
```

3:Start Services

```
sudo systemctl start nginx
```

```
sudo systemctl enable nginx
```

```
sudo systemctl start php-fpm
```

```
sudo systemctl enable php-fpm
```

```
sudo systemctl start mariadb
```

sudo systemctl enable mariadb

```

[ec2-user@ip-172-31-34-205 ~]$ sudo systemctl status nginx
CGroup: /system.slice/nginx.service
└─26228 "nginx: master process /usr/sbin/nginx"
└─26229 "nginx: worker process"
└─26230 "nginx: worker process"

Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Starting nginx.servi
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal nginx[26226]: nginx: the configu
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal nginx[26226]: nginx: configurati
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Started nginx.servi

[ec2-user@ip-172-31-34-205 ~]$ sudo systemctl status php-fpm
● php-fpm.service - The PHP FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; disabled; preset
   Active: active (running) since Wed 2026-04-22 14:04:27 UTC; 9min ago
     Main PID: 26225 (php-fpm)
    Status: "Processes active: 0, idle: 5, Requests: 0, slow: 0, Traffic: 0.0
      Tasks: 6 (limit: 1014)
     Memory: 11.2M
        CPU: 97ms
   CGroup: /system.slice/php-fpm.service
           └─26225 "php-fpm: master process (/etc/php-fpm.conf)"
             └─26231 "php-fpm: pool www"
               └─26232 "php-fpm: pool www"
                 └─26233 "php-fpm: pool www"
                   └─26234 "php-fpm: pool www"
                     └─26235 "php-fpm: pool www"

Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Starting php-fpm.serv
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Started php-fpm.serv
lines 1-18/18 (END) ...skipping...
● php-fpm.service - The PHP FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-04-22 14:04:27 UTC; 9min ago
     Main PID: 26225 (php-fpm)
    Status: "Processes active: 0, idle: 5, Requests: 0, slow: 0, Traffic: 0.00req/sec"
      Tasks: 6 (limit: 1014)
     Memory: 11.2M
        CPU: 97ms
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           └─26225 "php-fpm: master process (/etc/php-fpm.conf)"
             └─26231 "php-fpm: pool www"
               └─26232 "php-fpm: pool www"
                 └─26233 "php-fpm: pool www"
                   └─26234 "php-fpm: pool www"
                     └─26235 "php-fpm: pool www"

Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Starting php-fpm.service - The PHP FastCGI Process Manager...
Apr 22 14:04:27 ip-172-31-34-205.ec2.internal systemd[1]: Started php-fpm.service - The PHP FastCGI Process Manager.

```

4:check status

5:hit public ip we can see LEMP page



hello from LEMP.sh file

6:Setup Database Login:

mysql -u root -p

Create database: CREATE DATABASE FCT;

Create table:

USE FCT;

```
CREATE TABLE students ( id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100),email VARCHAR(50));
```

7: Create files

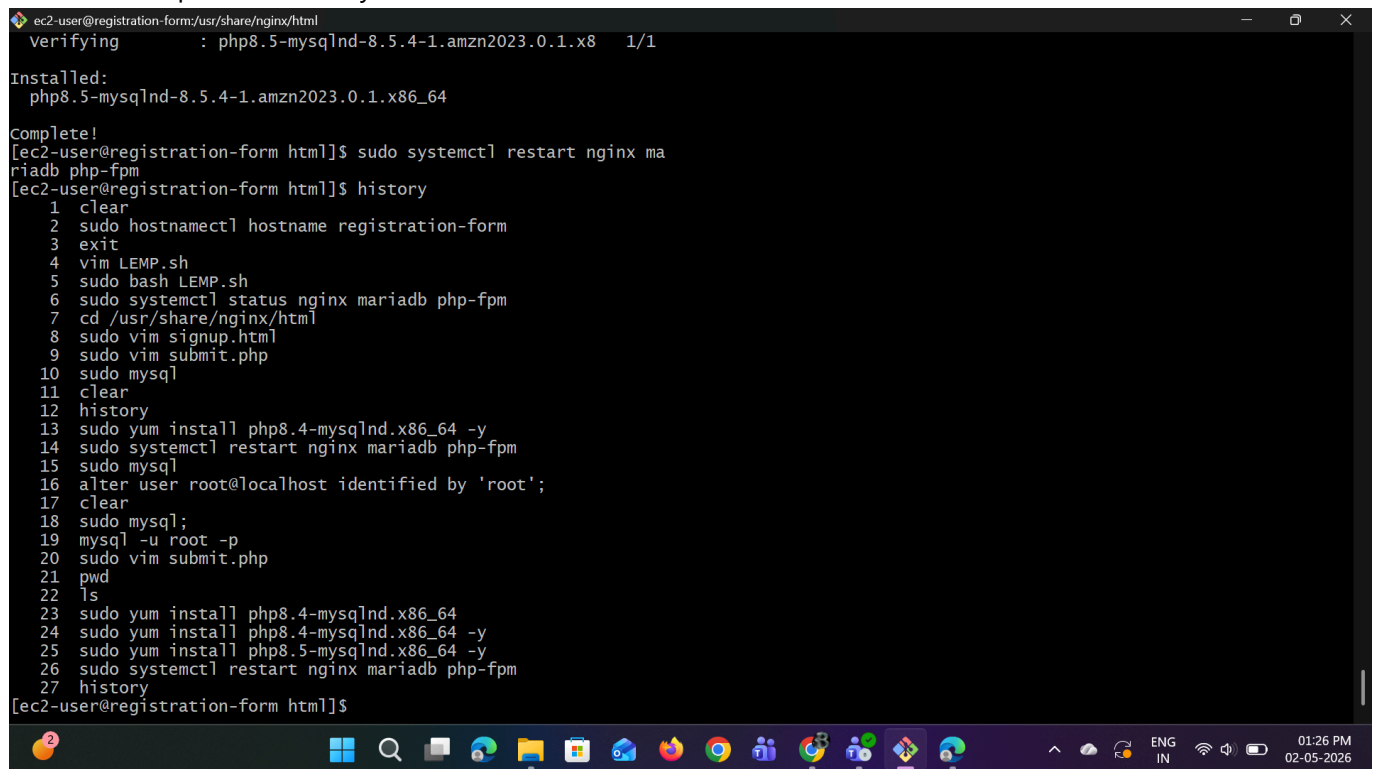
signup.html - for ui

submit.php -for backend

8: Download connector

```
sudo yum install php8.5-mysqlnd.x86_64 -y
```

9: After all steps restart the system

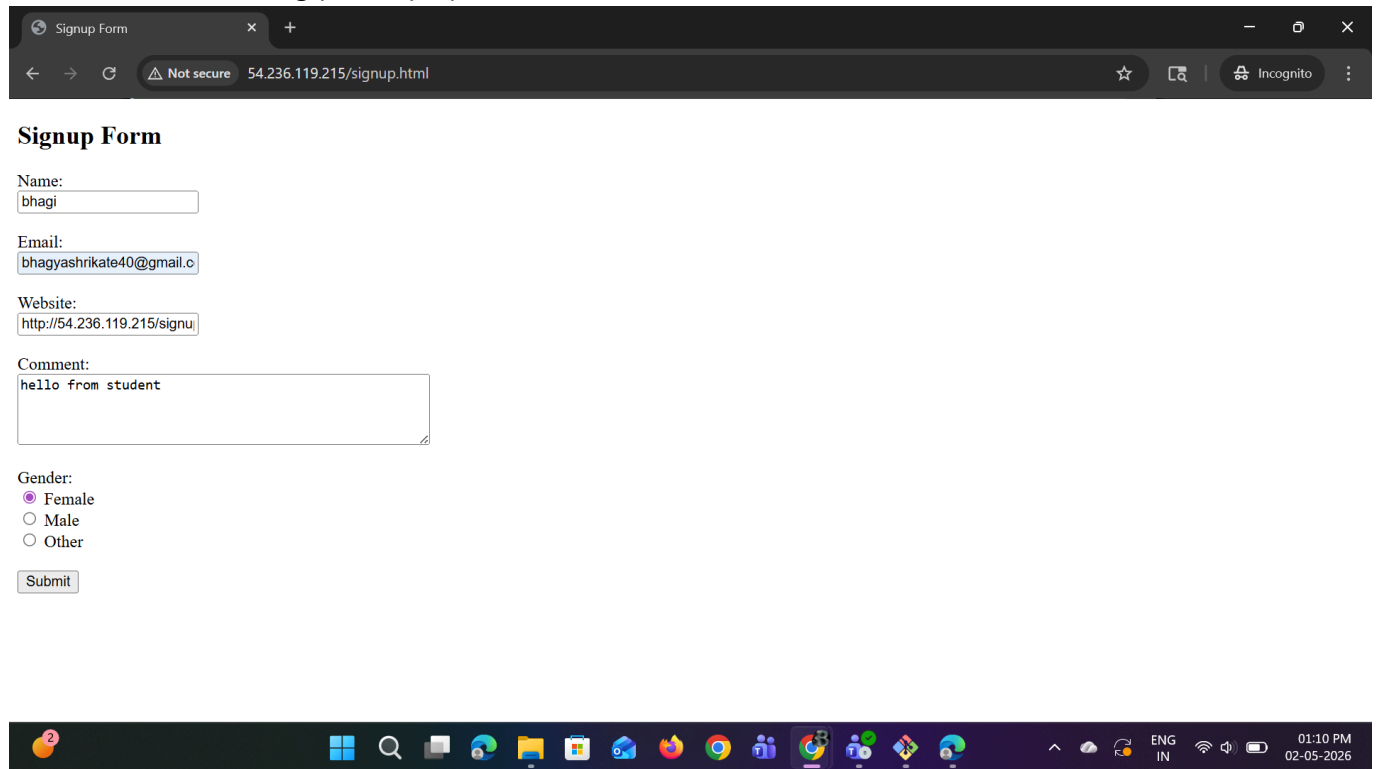
A terminal window titled 'ec2-user@registration-form:/usr/share/nginx/html' with standard window controls. The terminal output shows the verification of 'php8.5-mysqlnd-8.5.4-1.amzn2023.0.1.x86_64', followed by 'Installed:' and 'Complete!'. The user then runs 'sudo systemctl restart nginx mariadb php-fpm'. A 'history' command is executed, displaying a list of 27 commands including system setup, file editing, and package installation. The terminal ends with the prompt '[ec2-user@registration-form html]\$' and a Windows taskbar is visible at the bottom.

```
ec2-user@registration-form:/usr/share/nginx/html
Verifying : php8.5-mysqlnd-8.5.4-1.amzn2023.0.1.x86_64 1/1

Installed:
php8.5-mysqlnd-8.5.4-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@registration-form html]$ sudo systemctl restart nginx mariadb php-fpm
[ec2-user@registration-form html]$ history
 1 clear
 2 sudo hostnamectl hostname registration-form
 3 exit
 4 vim LEMP.sh
 5 sudo bash LEMP.sh
 6 sudo systemctl status nginx mariadb php-fpm
 7 cd /usr/share/nginx/html
 8 sudo vim signup.html
 9 sudo vim submit.php
10 sudo mysql
11 clear
12 history
13 sudo yum install php8.4-mysqlnd.x86_64 -y
14 sudo systemctl restart nginx mariadb php-fpm
15 sudo mysql
16 alter user root@localhost identified by 'root';
17 clear
18 sudo mysql;
19 mysql -u root -p
20 sudo vim submit.php
21 pwd
22 ls
23 sudo yum install php8.4-mysqlnd.x86_64
24 sudo yum install php8.4-mysqlnd.x86_64 -y
25 sudo yum install php8.5-mysqlnd.x86_64 -y
26 sudo systemctl restart nginx mariadb php-fpm
27 history
[ec2-user@registration-form html]$
```

10: Access website using public ip open browser



Signup Form

Name:
bhagi

Email:
bhagyashrikate40@gmail.com

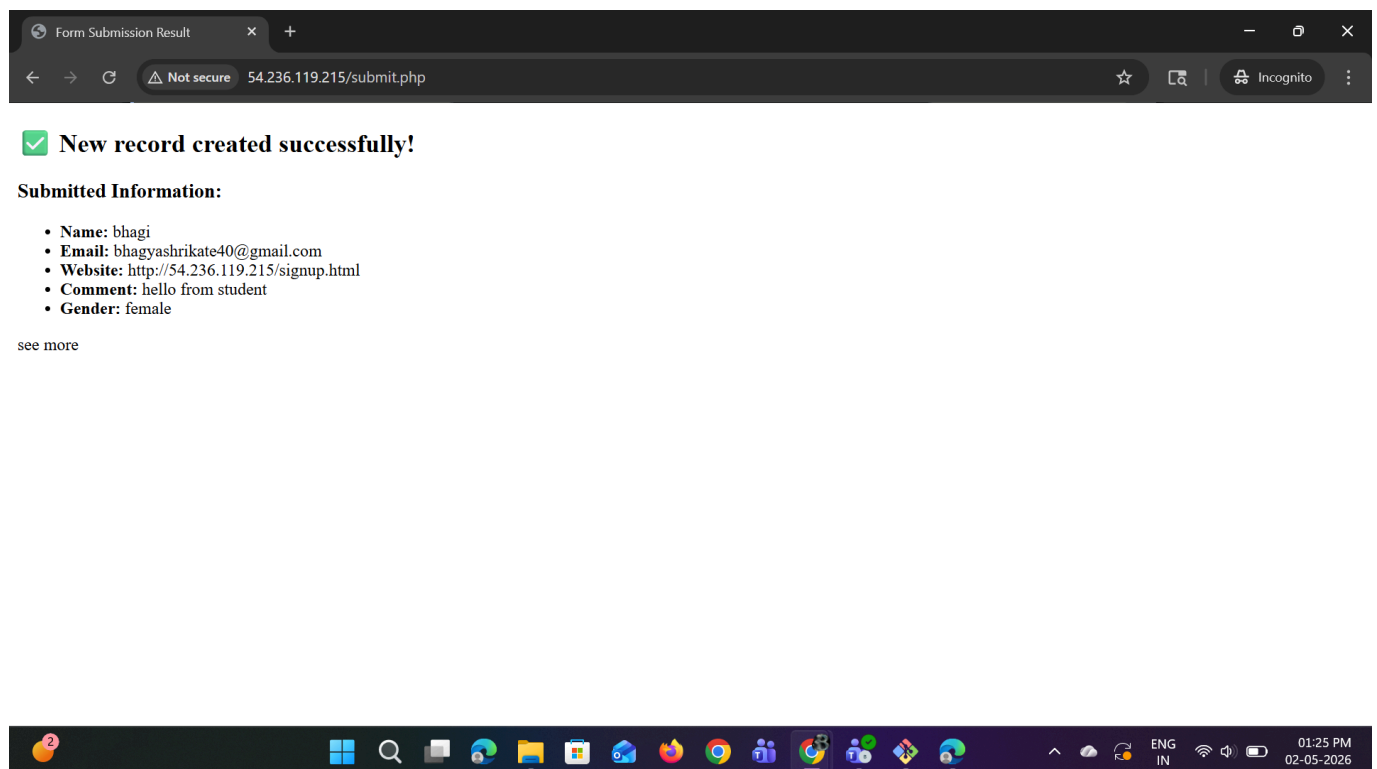
Website:
http://54.236.119.215/signup

Comment:
hello from student

Gender:
☒ Female
☐ Male
☐ Other

Submit

11: Fill form and submit data



Form Submission Result

Not secure 54.236.119.215/submit.php

✓ New record created successfully!

Submitted Information:

- Name: bhagi
- Email: bhagyashrikate40@gmail.com
- Website: http://54.236.119.215/signup.html
- Comment: hello from student
- Gender: female

see more

4) Summery

This project is a dynamic student registration form built using the LEMP stack. It shows how PHP and MySQL work together to collect and store user data. The website handles user input, processes it on the server, and saves it in the database. This is an important step in my journey towards DevOps, and cloud computing.